System calls

Exec.c

Input:

```
kaviya@localhost:~ — /usr/bin/vim exe.c

#include<stdio.h>
#include<unistd.h>
int main(){
        printf("Before exec()\n");
        execl("/bin/ls","ls",NULL);
        printf("After exec()\n");
        return 0;
}
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
"exe.c" 8L, 146B                                    8,1           All
```

Output:

```
kaviya@localhost:~

[kaviya@localhost ~]$ mkdir system call
[kaviya@localhost ~]$ mkdir systemcall
[kaviya@localhost ~]$ vi exe.c
[kaviya@localhost ~]$ vi exe.c
[kaviya@localhost ~]$ gcc exe.c
exe.c: In function 'main':
exe.c:7:9: error: expected declaration or statement at end of input
    7 |         return 0;
      |         ^~~~~~
[kaviya@localhost ~]$ vi exe.c
[kaviya@localhost ~]$ gcc exe.c
[kaviya@localhost ~]$ ./a.out
Before exec()
algorithm         bankers.c      Documents  kavi      sample.txt      Templates
a.out             call           Downloads  kavss     shellscripting  Videos
backup_test       cpu            exe.c      Music     system          while.sh
bankers           cpuscheduling  for.sh     Pictures  systemcall      XYZ
bankersalgorithm  Desktop        ifelse.sh  Public    systemcalls
[kaviya@localhost ~]$
```

Fork.c

Input:

```
kaviya@localhost:~/systemcalls — /usr/bin/vim fork.c

#include<stdio.h>
#include<unistd.h>
int main(){
        if(fork()==0){
                printf("This is CHILD process\n");
        }
        else{
                printf("This is PARENT process\n");
        }
        printf("PID: %d\n", getpid());
        return 0;
}
~
~
~
~
~
~
~
~
~
~
~
"fork.c" 12L, 198B                                    1,1           All
```

Output:

```
kaviya@localhost:~/systemcalls

[kaviya@localhost ~]$ cd systemcalls
[kaviya@localhost systemcalls]$ vi fork.c
[kaviya@localhost systemcalls]$ gcc fork.c
[kaviya@localhost systemcalls]$ ./a.out
This is PARENT process
PID: 2867
This is CHILD process
PID: 2868
[kaviya@localhost systemcalls]$
```

Opendir.c

Input:

```
kaviya@localhost:~/systemcalls — /usr/bin/vim opendir.c
#include<stdio.h>
#include<dirent.h>
int main(){
        DIR *dir;
        dir= opendir(".");
        if(dir==NULL){
                printf("Failed to open directory\n");
                return 1;
        }
        printf("Directory opened successfully!\n");
        closedir(dir);
        return 0;
}
~
~
~
~
~
~
~
~
~
~
"opendir.c" 13L, 225B                                   13,1          All
```

Output:

```
[kaviya@localhost systemcalls]$ vi fork.c
[kaviya@localhost systemcalls]$ vi opendir.c
[kaviya@localhost systemcalls]$ gcc opendir.c
[kaviya@localhost systemcalls]$ ./a.out
Directory opened successfully!
[kaviya@localhost systemcalls]$
```

Readdir.c

Input:

```
kaviya@localhost:~/systemcalls — /usr/bin/vim readdir.c
#include<stdio.h>
#include<dirent.h>
int main(){
        DIR *dir;
        struct dirent *entry;
        dir = opendir(".");
        if (dir==NULL){
                printf("Cannot open directory\n");
                return 1;
        }
        printf("Files in directory:\n");
        while((entry = readdir(dir)) != NULL){
                printf("%s\n", entry->d_name);
        }
        closedir(dir);
        return 0;
}
~
~
~
~
~
~
-- INSERT --                                    1,1           All
```

Output:

```
kaviya@localhost:~/systemcalls
PID: 2867
This is CHILD process
PID: 2868
[kaviya@localhost systemcalls]$ vi fork.c
[kaviya@localhost systemcalls]$ vi opendir.c
[kaviya@localhost systemcalls]$ gcc opendir.c
[kaviya@localhost systemcalls]$ ./a.out
Directory opened successfully!
[kaviya@localhost systemcalls]$ vi opendir.c
[kaviya@localhost systemcalls]$ vi readdir.c
[kaviya@localhost systemcalls]$ gcc readdir.c
[kaviya@localhost systemcalls]$ ./a.out
Files in directory:
.
..
exec.h
exec.c
pid.c
pidoutput
fork.c
opendir.c
readdir.c
a.out
[kaviya@localhost systemcalls]$
```

pid.c

input:

```
kaviya@localhost:~/systemcalls — /usr/bin/vim pid.c

#include<stdio.h>
#include<unistd.h>
int main(){
        printf("PID: %d\n", getpid());
        printf("PPID: %d\n", getppid());
        return 0;
}
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
"pid.c" 7L, 128B                                        1,1           All
```

output:

```
kaviya@localhost:~/systemcalls

[kaviya@localhost systemcalls]$ gcc opendir.c
[kaviya@localhost systemcalls]$ ./a.out
Directory opened successfully!
[kaviya@localhost systemcalls]$ vi opendir.c
[kaviya@localhost systemcalls]$ vi readdir.c
[kaviya@localhost systemcalls]$ gcc readdir.c
[kaviya@localhost systemcalls]$ ./a.out
Files in directory:
.
..
exec.h
exec.c
pid.c
pidoutput
fork.c
opendir.c
readdir.c
a.out
[kaviya@localhost systemcalls]$ vi pid.c
[kaviya@localhost systemcalls]$ gcc pid.c
[kaviya@localhost systemcalls]$ ./a.out
PID: 2962
PPID: 2537
[kaviya@localhost systemcalls]$
```